

Goal. Recap some highlights from the second half of Math 166.

1. Find the 9th order Maclaurin polynomial for $\int_0^x t \sin(t^2) dt$.
2. What type of curve is described by the equation in polar coordinates, $r = 9 \cos \theta$?
Convert the equation into rectangular coordinates.
3. What is the numerical value of the following series?

(a)
$$\sum_{n=0}^{\infty} \frac{3^n}{n!}$$

(b)
$$\sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n}}{(2n)!}$$

4. Consider the parametrization, $(x, y) = (t^2 - 3, 3t^2 + 9t + 1)$. At what speed is the particle moving at time $t = 2$?

5. Use Simpson's Rule with $n = 4$ to approximate the arclength of the curve $(x, y) = (t^2 - 3, 3t^2 + 9t + 1)$ from $t = 0$ to $t = 4$.